
SECOND YEAR GROUP PROJECT

PROPOSAL DEFENSE FEEDBACK

Project Action Plan

SmartAuto ERP - Automobile Spare Parts Distribution & Sales Management System

GROUP IS 24 — BSc Information Systems

GROUP MEMBERS

Index Number	Name of the Student	Assigned Primary Role
24021059	K. I. U. Thisera (Induwara)	Sales Representative / Mobile POS Lead
24021067	R. A. S. Thivanka (Sashik)	Store Manager / Inventory & Procurement Lead
24020451	L. A. C. R. Jayamali (Chathumi)	Business Owner / Admin & BI Reports Lead
24020771	V. Pavalaraj (Varshika)	Customer Portal & Public Services Lead

EXECUTIVE SUMMARY

During the proposal defense evaluation, the panel raised valid and critical concerns regarding the excessive scope, lack of domain specialization, and feasibility of developing a generic, multi-industry ERP across four undergraduate members within one academic year. In response, Group IS 24 has executed a complete strategic pivot: transitioning from an unmanageable multi-domain platform into SmartAuto ERP—a purpose-built Automobile Spare Parts Distribution and Sales Management System. Non-essential modules (such as comprehensive HR, payroll, and open-ended multi-industry handlers) have been removed. The system is strictly refocused on real-world auto parts workflows: inventory management, vehicle-to-part compatibility lookup, on-field mobile POS for sales representatives, and customer self-service ordering.

PANEL FEEDBACK SYNTHESIS & EVALUATION ANALYSIS

Panel Critique	Detailed Evaluation Context	Risk Identified by Panel
1. Overly Broad Scope & Feasibility	The panel noted that attempting a generic ERP across 4 unrelated domains (pharmacy, grocery, hardware, retail) with 10+ modules is an unfeasible “Sisyphean task” for a 4-member undergraduate team.	Critical Risk: Surface-level modules with negligible depth; guaranteed failure during code defense.
2. Reinventing the Wheel	Freely available, enterprise-grade open-source ERPs (e.g., ERPNext, Odoo) already exist for small businesses. Rebuilding a standard ERP without domain novelty lacks academic value.	High Risk: Inability to defend system uniqueness; potential suspicion of reusing existing solutions.
3. Vague Domain & Business Sizing	“Small and Medium Enterprises” was ill-defined without clear parameters regarding catalog complexity, employee count, or business operational boundaries.	Scope Ambiguity: Inability to establish realistic test cases or measurable user boundaries.
4. Superficial HR & Module Gaps	The panel observed that if an ERP claims to offer an HR module, tracking mere attendance is insufficient without complete payroll, performance reviews, and recruitment.	Design Flaw: Mislabeling basic operational tasks as full-scale enterprise modules.
5. Missing Client & Vague Needs	Without a real customer context, features were being conceptualized theoretically rather than solving an authentic, ground-level business bottleneck.	Practicality Gap: Developing disconnected features that fail to meet industry-specific realities.

COMPREHENSIVE ACTION TAKEN: THE STRATEGIC PIVOT

Following panel instructions, the group consulted supervisor Mr. Tharindu Ranathunga and restructured the project from the ground up. Below is the direct comparison of the changes adopted:

Previous Proposal (De-scoped)	SmartAuto ERP (Approved Pivot)
Domain: Generic, multi-industry ERP trying to cater to groceries, pharmacies, hardware, and retail outlets. Scope: 10+ broad modules (General HR, Payroll, Finance, SCM, Multi-tier Warehousing, Procurement). Target: Ambiguous SME sector without specific employee or transaction limits. Core Problem: Generic CRUD forms with superficial analytical capability and no domain edge.	Domain: Automobile Spare Parts Distribution and Sales Management System exclusively. Scope: 4 practical operational modules: Mobile POS, Spare Part Inventory, Administration/BI, and Customer Portal. Target: Mid-scale auto spare parts distributors, field sales representatives, auto repair workshops (B2B), and walk-in buyers (B2C). Core Novelty: Vehicle Compatibility Finder (Brand > Model > Year > Engine) and on-field Mobile POS.

VERTICAL WORKLOAD DISTRIBUTION ACROSS GROUP MEMBERS

The system is distributed vertically so that each member takes full end-to-end ownership of their assigned user role, user interface screens, business logic, and database entities:

Member 1: K. I. U. Thisera 24021059 Primary Role: Sales Representative Assigned Screens: Sales Rep Dashboard, Mobile POS Terminal, Order Management, Customer Management, Invoicing, Sales History. Core Responsibilities: Field sales processing via mobile POS, retail & walk-in sales checkout, customer registration, invoice generation, payment recording. Entities: customers, sales, sale_items, orders, order_items, payments.	Member 3: L. A. C. R. Jayamali 24020451 Primary Role: Business Owner / Administrator Assigned Screens: Authentication & RBAC, Owner Dashboard, Employee Accounts, Business Reports, Activity Logs, System Settings. Core Responsibilities: User authentication & role security, employee sales tracking, profit/sales/stock analytics, audit trails, system configuration. Entities: users, roles, employees, notifications, activity_logs, settings.
Member 2: R. A. S. Thivanka 24021067 Primary Role: Store Manager Assigned Screens: Product Management, Inventory & Stock Movements, Purchase Orders, Supplier Management, Spare Part Finder Logic. Core Responsibilities: Spare parts cataloging, stock in/out adjustments, low-stock threshold detection, vehicle compatibility matrix, supplier procurement. Entities: products, categories, brands, vehicles, compatibility, inventory, suppliers, purchase_orders.	Member 4: V. Pavalaraj 24020771 Primary Role: Shop Customer/Public User Assigned Screens: Public Product Catalog, Vehicle-Based Spare Part Finder (Customer UI), Online Ordering Portal, Order Tracking, Delivery Status. Core Responsibilities: Customer part search by vehicle make/model, self-service order placement for repair shops, order delivery tracking. Entities: product_catalog, customer_orders, deliveries, compatibility (view).

EXPLICIT OUT-OF-SCOPE BOUNDARIES

To ensure strict feasibility within the undergraduate timeframe, the following features are confirmed as out of scope: (1) Integrated online credit card payment gateways; (2) Live GPS route tracking for sales reps; (3) Complex recruitment and payroll HR; (4) Offline sync; and (5) Multi-warehouse distribution.